LAB 7: Fuzzing for Input Validation Bugs

Attendance Eligibility Parser — Submission Report

Objective. Apply property-based fuzz testing with Hypothesis to discover edge-case failures in three input-parsing functions, then harden the functions against malformed values.

Repository and filenames. The instructor's LAB7 folder specifies *attendance_eligibility_parser.py* and *test_attendance_eligibility_parser.py*. These names are used throughout this submission.

1. Functions and defects addressed

normalize_srn(srn) — the starter assumed a non-empty string. Non-string values such as None caused an AttributeError. The fixed function checks the type, removes all whitespace, uppercases valid strings, and returns an empty string for invalid types.

attendance_from_ratio(text) — the starter assumed exactly one slash, integer values, and a non-zero denominator. Malformed ratios could raise ValueError or unpacking errors, and a zero denominator caused ZeroDivisionError. The fixed function validates structure and integer conversion, rejects held values at or below zero and attended values outside 0..held, and returns None for invalid input.

attendance_from_register(marks) — the starter assumed a non-empty string containing only P/A marks. Empty values could divide by zero; invalid types could fail on strip; other characters distorted the percentage. The fixed function validates type, non-empty content, and allowed characters before calculating.

2. Hypothesis fuzz tests

The updated test file defines one property-based test for each function. Each test uses `st.text() | st.none()`, so Hypothesis explores arbitrary Unicode strings and None. The property checked is that each function handles the generated value without raising an exception. Hypothesis automatically shrinks a failure to a small counterexample.

3. Required terminal screenshots

The following four screenshots are required by the instructor's README. They must be captured from the actual local run. They are deliberately left as capture slots rather than fabricated terminal evidence.

SS1 — Original buggy application

Run the starter version before replacing it. Show the command and sample inputs/output.

```
[binduphilip@Shanas-MacBook-Air LAB7_buggy % ls                                ]
attendance_eligibility_parser.py        test_attendance_eligibility_parser.py
[binduphilip@Shanas-MacBook-Air LAB7_buggy % python3 -m pytest test_attendance_eligibil]
ity_parser.py
================================= test session starts ==================================
platform darwin -- Python 3.9.6, pytest-8.4.2, pluggy-1.6.0
rootdir: /Users/binduphilip/Downloads/LAB7_buggy
plugins: hypothesis-6.141.1
collected 3 items

test_attendance_eligibility_parser.py ...                                      [100%]

================================== 3 passed in 1.93s ===================================
```

SS2 — Failing fuzz tests

Run the completed Hypothesis tests against the original buggy module. Show the failure output and minimized example.



SS3 — Fixed application

Run the fixed parser with representative valid sample inputs and show its output.

```
binduphilip@Shanas-MacBook-Air LAB7_buggy % cd ~/Downloads/LAB7
binduphilip@Shanas-MacBook-Air LAB7 % python3 attendance_eligibility_parser.py
==== Running normalize_srn ====
Enter an SRN (e.g. pes1ug23cs001): pes1ug24cs427
Normalized SRN: PES1UG24CS427

==== Running attendance_from_ratio ====
Enter classes attended and held as ATTENDED/HELD (e.g. 30/40): 38/40
Attendance percent: 95

==== Running attendance_from_register ====
Enter the register as P/A marks (e.g. PPAPP): PPAPP
Attendance percent: 80
```

SS4 — Passing fuzz tests

Run pytest against the fixed module and show the final passing summary.

```
binduphilip@Shanas-MacBook-Air LAB7 % python3 -m pytest -v test_attendance_eligibility_parser.]
py
======================================== test session starts ========================================
platform darwin -- Python 3.9.6, pytest-8.4.2, pluggy-1.6.0 -- /Library/Developer/CommandLineT
ools/usr/bin/python3
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: /Users/binduphilip/Downloads/LAB7
plugins: hypothesis-6.141.1
collected 3 items

test_attendance_eligibility_parser.py::test_normalize_srn_no_crash PASSED                  [ 33%]
test_attendance_eligibility_parser.py::test_attendance_from_ratio_no_crash PASSED          [ 66%]
test_attendance_eligibility_parser.py::test_attendance_from_register_no_crash PASSED       [100%]

========================================= 3 passed in 0.49s =========================================
```

4. Reflection

What is fuzz testing and why is it useful?
Fuzz testing supplies automatically generated, unexpected inputs to a program. It complements hand-written examples by exploring unusual strings, Unicode, empty values, and invalid types that a developer may not think to test manually.

How did Hypothesis help identify edge cases?
The @given decorator and text-or-None strategy allow Hypothesis to generate many inputs automatically. When a test fails, Hypothesis reduces the input to a simpler counterexample, making the underlying validation defect easier to understand.

How were the functions made more robust?
Each function now validates the input type and format before processing. Ratio calculations guard against malformed values, zero/negative denominators, and impossible attendance counts. Register calculations reject empty strings and characters other than P/A. Invalid data returns a documented safe result instead of crashing.

5. Commands to capture evidence on macOS

```
cd /path/to/SE-Labs-PES1UG24CS427/LAB7
python3 -m pip install -r requirements.txt

# SS3: run the fixed application
python3 attendance_eligibility_parser.py
# Enter: pes1ug23cs001
# Enter: 30/40
# Enter: PPAPP

# SS4: run the passing fuzz tests
python3 -m pytest -q test_attendance_eligibility_parser.py
```

Important note about SS1 and SS2: these must be captured before overwriting the starter file with the fixed implementation. If the original starter is no longer present locally, retrieve it from the instructor's commit and run the tests against that copy first.

Source limitation: the supplied four-page handout describes the steps and deliverables but does not print the three reflection questions. The reflections above answer the three learning themes supported by the handout; replace them with exact question wording if your instructor provides a separate question sheet.